Software Engineering

Test Case Template and Examples

Test Case Template (Doc:T_01)

Test Case #:	**Test Case Name:**	**Page: 1 of ..**
System:	**Subsystem:**	
Designed by:	**Design Date:**	
Executed by:	**Execution Date:**	
Short Description:		

Pre-conditions

Step	Action	Expected System Response	Pass/ Fail	Comment
1				
2				

Post-conditions

Test Case Example1 (simple test)

Test Case #: 2.2 **Test Case Name:** Change PIN **Page:** **1 of 1**

System: **ATM** **Subsystem:** **PIN**

Designed by: **ABC** **Design Date:** **28/11/2004**

Executed by: **Execution Date:**

Short Description: **Test the ATM Change PIN service**

Pre-conditions
The user has a valid ATM card - The user has accessed the ATM by placing his ATM card in the machine The current PIN is 1234
The system displays the main menu

Step	Action	Expected System Response	Pass/ Fail	Comment
1	Click the 'Change PIN' button	The system displays a message asking the user to enter the new PIN		
2	Enter '5555'	The system displays a message asking the user to confirm (re-enter) the new PIN		
3	Re-enter '5555'	The system displays a message of successful operation The system asks the user if he wants to perform other operations		
4	Click 'YES' button	The system displays the main menu		
5	**Check post-condition 1**			

Post-conditions
1. The new PIN '5555' is saved in the database

Test Case Example2 (complex test may be divided into more than test cases)

Test Case #: 2.3 **Test Case Name:** Change PIN **Page:** 1 of 1

System: ATM **Subsystem:** PIN

Designed by: ABC **Design Date:** 28/11/2004

Executed by: **Execution Date:**

Short Description: Test the ATM Change PIN service

Pre-conditions
The user has a valid ATM card - The user has accessed the ATM by placing his ATM card in the machine The current PIN is 1234
The system displays the main menu

Step	Action	Expected System Response	Pass/ Fail	Comment
1	Click the 'Change PIN' button	The system displays a message asking the user to enter the new PIN		
2	Enter '5555'	The system displays a message asking the user to confirm (re-enter) the new PIN		
3	Re-enter '5555'	The system displays a message of successful operation The system asks the user if he wants to perform other operations		
4	Click 'YES' button	The system displays the main menu		
5	**Check post-condition 1**			
6	Repeat steps 1,2,3 using another PIN say '6666' and click 'NO' button	The system is exited and displays a greeting message asking the user to place his ATM card in the machine		
7	**Check post-condition 2**			
8	Repeat steps 1,2, using another PIN say '7777'	The system displays a message asking the user to confirm (re-enter) the new PIN		
9	Enter a wrong confirmation (say '9876')	The system displays a message of **unsuccessful** operation and asks the user to confirm the correct PIN		
10	Re-enter '7777'	The system displays a message of successful operation The system asks the user if he wants to perform other operations		
	Click 'NO' button	The system is exited and displays a greeting message		
	Check post-condition 3			

Post-conditions

1. The new PIN '5555' is saved in the database
2. The new PIN '6666' is saved in the database
3. The new PIN '7777' is saved in the database

Test happy path
Test also failure condition (step 9)

Frontend UI Test Case 1.1

Test Case #: 1.1 **Test Case Name: Load User's Location**
Page: 1 of ..

System: Lost Grenadiers UI **Subsystem: Routing**

Designed by: Audrey Dickson **Design Date:**

Executed by: **Execution Date:**

Short Description: Verify that the user's correct location loads correctly.

Pre-conditions
- user has cellular data/internet access
- user opens supported browser
- user navigates to the Lost Grenadiers website

Step	Action	Expected System Response	Pass/ Fail	Comment
1				
2				

Post-conditions

Frontend UI Test Case 1.2

Test Case #: 1.2 **Test Case Name: Show Route Preview – Happy Path**
Page: 1 of ..

System: Lost Grenadiers UI **Subsystem: Routing**

Designed by: Audrey Dickson **Design Date:**

Executed by: **Execution Date:**

Short Description: Verify that a route preview is generated when user selects a valid destination.

Pre-conditions
- map is loaded
- user location is available
- routing backend is connected

Step	Action	Expected System Response	Pass/ Fail	Comment
1				
2				

Post-conditions

Frontend UI Test Case 1.3

Test Case #: 1.3 **Test Case Name: Show Route Preview – Failure Condition**
Page: 1 of ..

System: Lost Grenadiers UI **Subsystem: Routing**

Designed by: Audrey Dickson **Design Date:**

Executed by: **Execution Date:**

Short Description: Verify system behavior when routing fails or invalid input is entered.

Pre-conditions
- map is loaded
- user location is available
- routing backend is connected

Step	Action	Expected System Response	Pass/ Fail	Comment
1				
2				

Post-conditions

Frontend UI Test Case 1.4

Test Case #: 1.4 **Test Case Name: Indoor Mode Overlay Toggle**
Page: 1 of ..

System: Lost Grenadiers UI **Subsystem:**

Designed by: **Design Date:**

Executed by: **Execution Date:**

Short Description: Verify that indoor mode shows floor overlays correctly.

Pre-conditions
- map is loaded
- a building is selected
- overlay image files exist

Step	Action	Expected System Response	Pass/ Fail	Comment
1				
2				

Post-conditions

Frontend UI Test Case 1.5

Test Case #: 1.5 **Test Case Name: Show Route – Happy Path**
Page: 1 of ..

System: Lost Grenadiers UI **Subsystem: Routing**

Designed by: **Design Date:**

Executed by: **Execution Date:**

Short Description: Verify that a route preview is generated when user selects a valid destination.

Pre-conditions
- map is loaded
- user location is available
- routing backend is connected

Step	Action	Expected System Response	Pass/ Fail	Comment
1				
2				

Post-conditions

Frontend UI Test Case 1.6

Test Case #: 1.6 **Test Case Name: Show Route – Failure Condition**
Page: 1 of ..

System: Lost Grenadiers UI **Subsystem: Routing**

Designed by: **Design Date:**

Executed by: **Execution Date:**

Short Description: Verify system behavior when routing fails or invalid input is entered.

Pre-conditions
- map is loaded
- user location is available
- routing backend is connected

Step	Action	Expected System Response	Pass/ Fail	Comment
1				
2				

Post-conditions

Backend API Test Case 2.1

Test Case #: 2.1 **Test Case Name:** Valid Routing API Request **Page:** 1 of 1

System: Lost Grenadiers Backend **Subsystem:** Flask API

Designed by: Andrew Swartz **Design Date:** 2/18/2026

Executed by: **Execution Date:**

Short Description: Test the ATM Change PIN service

Pre-conditions
- Flask server is running
- PostgreSQL + pgRouting is running
- Valid nodes exist in the database

Step	Action	Expected System Response	Pass/ Fail	Comment
1	Click the 'Change PIN' button	The system displays a message asking the user to enter the new PIN		
2	Enter '5555'	The system displays a message asking the user to confirm (re-enter) the new PIN		
3	Re-enter '5555'	The system displays a message of successful operation The system asks the user if he wants to perform other operations		
4	Click 'YES' button	The system displays the main menu		

Post-conditions

1. The new PIN '5555' is saved in the database